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RESEARCH ARTICLE

Electric Vehicles CAN Bus Cyber Attacks Detection Using Adaptive Neuro Fuzzy Inference System

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ABSTRACT The Electric Vehicle (EV) industry has recently experienced notable technological progress in the field of Controller Area Network (CAN) protocol. The use of CAN bus protocol in EVs is exposed to intrinsic cybersecurity risks and consequently causing EV damages as a result of lack of authentication, authorization, and accounting mechanisms. This paper examines the vulnerabilities within the EVs' CAN bus protocol and explores potential strategies for mitigating cyber threats (i.e. Denial of Service (DOS) and impersonation attacks). In particular, the paper proposes Adaptive Neuro Fuzzy Inference System (ANFIS) based detection techniques superimposed with Subtractive Clustering (SC) and Fuzzy C-Means clustering (FCM). Results demonstrate that the proposed ANFIS-SC and ANFIS-FCM detection model testing accuracy is 99.6%, TPR and TNR values are above 99.8%. In addition to the low FPR and FNR values are less than 0.2% of the proposed ANFIS-SC and ANFIS-FCM detection techniques. The overall F1 score is above 98.8%.

INDEX TERMS Adaptive neuro fuzzy inference system (ANFIS), cyber-attacks detection, EV CAN bus, fuzzy C-means clustering (FCM).

I. INTRODUCTION

Electric Vehicles (EVs) are trending as technological advances to overcome global warming and limited fossil fuel resources. Based on a report from the International Energy Agency (IEA), in 2030, around 30 million EVs will be operational on the roads. Many researchers in recent years have studied the implementation of deep learning and deep reinforcement learning in the transportation system to achieve autonomous and intelligent driving [1], [2], [3], [4], [5], which brought up the cybersecurity concern of such systems. Despite EV's potential to address global concerns and anticipated growth rate, their adaptation faces many challenges, including the availability of adequate charging infrastructure, range anxiety [6], and safety. The Electronic Control Units (EUCs) within an EV are responsible for controlling EVs' functionality. Such units communicate through the Controller Area Network (CAN) bus protocol for effective operation. Nevertheless, EV occupants' safety is a pressing need as the CAN bus is vulnerable to cyber-attacks [7]. When messages are transferred among ECUs, they are transferred with unique CAN bus IDs and data field values (See Table 1. 1 for CAN bus message format in 11-bit mode alongside its elements and their definitions). The three key objectives of a cybersecurity study are described by the Confidentiality, Integrity, and Availability (CIA) model. [8]. Using the Authentication, Authorization, and Accounting (AAA) model, the cybersecurity goals described by the CIA model can be achieved [8].

The electric vehicles are promising technologies with a variety of applications. The literature has investigated their energy storage [9], [10], [11], [12], energy management and infrastructure [13], [14], [15], [16], [17], autonomy [18], [19], power electronics [20], grid support [21],

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TABLE 1. CAN bus message format in 11-bit mode.

1 bit	11 bits	4 bits	0 to 8 bytes	16 bits	2 bits	7 bits
SOF	CAN ID	DLC	Data Field	CRC	ACK	EOF
The Start Of Frame (SOF) bit is for alerting all ECUs in the EV that a CAN message transmission has started	CAN ID is used to identify which ECU should receive the message. The CAN bus protocol assigns priorities to each message, and it is determined by the CAN ID value. The lower the value of the CAN ID, the higher the priority of the message is.	DLC determines the byte length of the data field. The values of the DLC field are between 0 and 8.	The data Field includes the payload data that is received and interpreted by the intended ECUs.	CRC Field is used to detect message transfer errors,	The acknowledge (ACK) Field is used to confirm that the intended ECUs receive the proper CAN message.	End Of Frame (EOF) field is a sign that the CAN message has ended

load forecasting [22], economics [23], motor health [24]. Cybersecurity in EV control networks is a challenge for researchers; there are many studies that have covered the cybersecurity of electric vehicles and electric charging stations (EVCSs), where they discussed cyberattacks, their vectors, and some defense mechanisms against such attacks.

In the literature [25], various hybrid anomaly detection methods are used for flood cyberattacks in the CAN bus that are based on wavelet transformation and Support Vector Machines (SVM). The parameters of the anomaly detection method are tuned using a Modified Crow Search Algorithm (MCSA) due to the high complexity of the EV CAN bus data. In [26], a machine learning-based cyber-attack detection scheme was developed for CAN bus data inside EVs. The machine learning algorithm implemented is a one-class SVM that utilizes a modified version of the bat algorithm to optimize its performance. The authors are clear that the effective input features may influence the model's accuracy. The model training time of the proposed algorithm is high; if the meaningful features are considered, then the training time may be reduced. The explainable AI (XAI)-based learning algorithm has been developed for intrusion detection [27]. In [28], researchers implemented a Generative Adversarial Network (GAN) to detect cyber attacks on the CAN bus of EVs. Since the optimization process of the GANs is complex, a modified Firefly algorithm was used. The authors in [29] applied the same concept as the previous paper [28], where they used an anomaly detection method of cyber-attacks in the CAN bus of EVs based on a modified SVM. The difference here is that they used a Modified Social Spider Algorithm (MSPA) instead of the MCSA to fine-tune the parameters of the SVM model to detect DOS attacks. In [30], the authors focused on detecting cyber-attacks that target EVs during the charging. They implemented two approaches for attack detection; one is the static detector, and the other is the dynamic detector. The simulations showed that the dynamic detector is superior to the static detector in attack detection. However, in the developed framework, the authors do not include nonlinearities in the battery model and attacks on electric vehicle supply equipment.

The researchers in [31] studied the complex cyber-attacks that target the Energy Management systems (EMS) in Hybrid Electric Vehicles (HEVs). A detection technique based on probability is developed to improve the cyber security of HEVs. The limitation of this approach lies in its real-time implementation within an in-vehicle monitoring system. Therefore, a dual-defense approach—combining information security and control security is required to mitigate cyber threats and support the development of real-time cybersecurity. In [32], the authors proposed a cyber-attack detection method in the CAN bus of EVs. They used Artificial Neural Networks (ANN) and Multilayer Perceptron (MLP) to detect attacks such as DOS, Fuzzy attacks, and other attacks that target several components inside the EV. The authors in [7] implemented LSTM as an Intrusion Detection System (IDS) to detect cyber-attacks against the CAN bus in EVs. The researchers in [33] analyzed various cyber-attack scenarios that can affect the onboard charging system inside EVs. They concluded that cyber-attack damage to the equipment inside the EV charging system can be minimized completely if cyber-attack prevention measures are considered while designing the EV charging system.

In [34], a novel intrusion detection system (IDS) mechanism is developed for cyber-attacks on the EV CAN bus. The detection mechanism is based on the offset ratio and the time period between the sending and receiving of a CAN message. The developed IDS mechanism was applied to the CAN bus under DOS, fuzzy, and impersonation attacks, where it performs well in detecting such attacks. The proposed model needs to be validated on the other in-vehicle communication protocols, i.e., FlexRay, LIN protocol, and Ethernet. This ensemble learning framework has not been validated for different types of attacks. Consequently, the proposed framework is unable to effectively mitigate attack data when multiple attacks occur simultaneously.

Since EVs charge and discharge through the EV Charging Stations (EVCSs), the cybersecurity of EVCSs needs to be studied as well. The Open Charge Point Protocol (OCPP) is a protocol that provides standards for communication data between the Central Server (CS) and the Charging Point (CP) of EVs [35]. The authors in [35] studied the Man in the Middle attacks that target the communication data between CS and CP and developed a protection scheme against this type of attack. These strategies may not be sufficient to secure communication channels with transport layer security. It is required to develop VPNs between all peers in the

vehicle infrastructure and implement firewalls and false data detection systems to generate a trusted environment for operators and users, ultimately promoting the adoption of these technologies.

In [36], the researchers studied the potential of a remote eavesdropping attack on the charging messages of 54 EVCSs with Combined Charging System (CCS) connectors. The authors in [37] proposed a cybersecurity solution to the OCPP by using authentication trees that allow the removal of data from collections while achieving end-to-end security and non-repudiation. In [38], the researchers found that the vulnerabilities present in the OCPP can result in spoofing, stealing, or altering sensitive EV charging data such as the EVCS identifiers. In [39], Schneider Electric found security flaws in the EV-Link chargers. A remote attacker can bypass credentials that are hard-coded and perform code and SQL injections to disable the charging and gain access to the web interface. In [40], vulnerabilities alongside attack mitigation techniques were discussed for CirCarLife, which is an EVCS. They found that attackers could bypass authentication through a different path, allowing them to know credentials permitting access to sensitive information. In [41], Kaspersky Lab explained the flaws of the Charge Point Home smartphone application. The flaws facilitated bypassing the users' authentications, imitating them, and altering the EV charging data. In [42], the cybersecurity of EVCS charging practices has been discussed; Radio Frequency Identification (RFID) cards and QR codes are the main things that are used in authenticating the user in public EVCSs where they are prone to cyber-attacks.

Based on the conducted literature review, one can notice that the cyber-attacks and defense studies of the EVCSs are all related to the weak cybersecurity practices adopted in the protocols, web or smartphone applications, access cards, etc., and that improving the security practices of the used protocols, applications, or cards is the solution to almost all cyber security concerns. However, this is not the case for the EV attacks and defense studies since the problems are varied and more complex. The security problems in EVs are not limited to the protocols adopted but are complex and deep-reaching even in the EV charging system. Based on the conducted literature review, one can observe that the dominant attack vector for cyber-attacks is the EVs CAN bus. This is due to the poor adoption of cyber security practices in the EVs CAN bus protocol and thus being the most vulnerable threat for cyber-attacks.

In a nutshell, most studies in the literature focused on the anomaly detection of cyber-attacks in the EVs CAN bus messages, and there are only a few studies adopting the classification approach of cyber-attacks. Therefore, developing efficient detection techniques that can detect and classify the cyber-attacks in the EV CAN bus to provide a defense mechanism against such attacks is necessary. Moreover, implementing over-sampling techniques on the EV CAN bus messages under cyber-attacks to improve the performance of the detection techniques is another gap observed in the literature.

This study considers the battery SoC model node and CAN bus unit for the cyber attack analysis. The Battery SoC model is an approach for obtaining the behavior of battery output in various external and internal environments. The historical data of the SoC model is utilized to predict the battery state at any time for checking battery health. Additionally, the model is utilized to estimate the distance of EVs until the battery will not reach its minimum SoC level. The traveled distance information is taken from stored accessible data on the CAN bus by sensors. For EVs, the battery is the only source; hence, the precision of the level calculation is more important. If there is a cyberattack on the battery model information, then the information transfer by the CAN bus is mismatched, and the decision logic unit may make the wrong decision. Battery management systems also perform mail operations. This paper aims to develop cyber-attack detection techniques to detect the cyber-attacks targeting the EVs CAN bus and mitigate them. Detection techniques based on the Adaptive Neuro-Fuzzy Inference System (ANFIS) algorithm will be proposed and developed to detect and classify cyber-attacks. The performance of the ANFIS algorithm-based detection techniques will be compared with that of other machine learning and deep learning-based techniques. Synthetic Minority Oversampling Technique (SMOTE) will be implemented on the EV CAN bus messages under cyber-attacks that are used to develop the detection techniques, to improve the performance of the detection techniques, and to better classify the attacks.

This paper is structured as follows: Section II discusses the cyber-attacks under investigation. Section III presents the implementation of the EV CAN bus cyber-attack detection techniques, starting with an overview of the ANFIS algorithm and moving forward to the integration of SC and FCM into the ANFIS algorithm. Section IV discusses the performance evaluation of the implemented detection techniques. Finally, section IV concludes the paper and suggests future research directions.

II. CAN COMMUNICATION NETWORK

Charging stations play a crucial role in the electric vehicle (EV) industry by supplying power, either AC or DC, based on the vehicle's requirements. The infrastructure of a charging station comprises both hardware and software components that facilitate energy transfer while ensuring safety. Effective communication is essential for transmitting messages between the battery management system and the charger. The EV internal communication network is shown in Fig. 1.

The CAN buses are the main part of the EVs. CAN bus provides communication among all other sections of EV. It is a serial network system for the EVs sector. The electronic control units (ECU) are embedded via the CAN bus, which provides a central networking system.

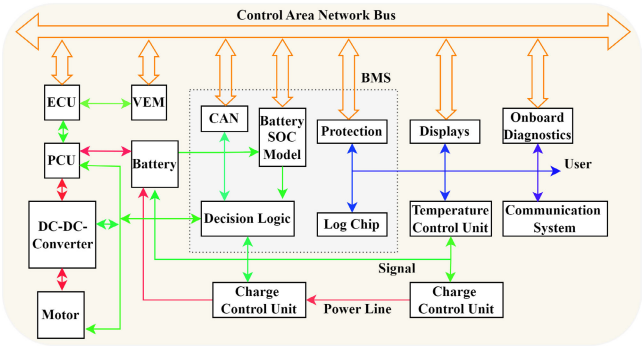


FIGURE 1. EV internal communication network.

The ECU and CAN relate to the complete system via software communication without allowing device overload. CAN delivers rapid communication between ECUs, real-time sensing, and control systems. The EV data, including battery level, engine temperature, torque, oil level, speed of the transmission system, and other information from different sections, are connected via the CAN interface. CAN is a high-speed, two-wire, half-duplex communication system known for its flexibility and reliability compared to traditional serial interfaces like RS232. It helps eliminate the need for complex and expensive dual-port infrastructures. While CAN may not be the best choice for systems requiring minimal network management or handling data packets larger than 8 bytes, its functionality can be extended using higher-layer protocols. These protocols integrate additional software with the CAN physical layer, enabling advanced network features such as extended message lengths and improved network management through node IDs. The SAE J1939 standard enhances CAN communication by introducing a software-based higher-layer protocol. It supports node addressing and allows for data transmission beyond the standard 8-byte limit, handling messages up to 1785 bytes. Additionally, CAN transmits up to 1765 bytes of data using a transport protocol (TP). For effective CAN-based communication between a vehicle and a charging system, a dedicated data circuit is required. This circuit facilitates the exchange of charge control parameters, including current setpoints, voltage and current measurements, and status flags for charging and vehicle conditions.

However, the CAN bus lacks built-in security measures such as authentication, encryption, and message verification. This vulnerability makes it more susceptible to attacks compared to protocols like Transmission Control Protocol/Internet Protocol (TCP/IP). Without proper authentication, attackers can exploit the network by injecting malicious messages or executing replay attacks to compromise an ECU. However, advancements in data mining techniques have provided solutions to mitigate such threats. Researchers have developed methods to detect and filter out abnormal communication traffic, enhancing the security of CAN-based systems.

III. CYBER-ATTACKS UNDER INVESTIGATION

This section presents and discusses the cyberattacks under investigation. Two types of cyber-attacks (i.e. Denial of Service (DOS) attack and impersonation attack) that target the EV CAN bus are considered in this paper as described below, while the CIA classification of these considered EV CAN bus cyber-attacks is shown in Table 1.

TABLE 2. CIA classification of the considered EV CAN bus cyber-attacks.

CAN bus cyber-attack	CIA classification
DOS	Availability
Impersonation	Confidentiality and Integrity

- Denial of Service (DOS) attack: This type of attack is done when an attacker sends high-frequency, priority data in the CAN bus, as shown in Fig. 3, causing traffic inside the bus and an interruption of communication. In this paper, the DOS attack objective is to delay or obstruct the information exchange between the nodes in the EV CAN bus and, thus, target the availability aspect of the CAN bus messages.
- Impersonation attack: The impersonation attack illustrated in Fig. 4 is an attack that stops a targeted ECU node from transmitting data and impersonates the node by sending messages with the same CAN ID as the targeted node but with different data field values. In this paper, the impersonation attack aims to access the confidential information of the EV CAN bus and impersonate an ECU node in the EV CAN bus while altering the CAN bus messages sent from the impersonated node. The impersonation attack thus targets the Confidentiality and Integrity aspects of the CAN bus messages.

This section proposes new algorithms for the CAN bus cyber-attacks detection based on the hybrid combination of Adaptive Neuro-Fuzzy Inference System (ANFIS) is a deep learning-based algorithm with subtractive clustering (SC) and fuzzy C-means clustering (FCM) being unsupervised machine learning algorithms. The proposed algorithms are the ANFIS-SC and the ANFIS-FCM. A raw public dataset of an EV CAN bus injected with the DOS and impersonation attacks will be imported. Then, data preprocessing techniques, such as labeling and filling in the missing values, were applied to the raw public dataset to prepare the data for the ANFIS-SC and ANFIS-FCM algorithms training and testing. The proposed techniques' performances will be compared with other machine learning and deep learning-based techniques for the detection of CAN bus cyber-attacks.

A. ADAPTIVE NEURO-FUZZY INFERENCE SYSTEM (ANFIS) ARCHITECTURE

The Adaptive Neuro-Fuzzy Inference System (ANFIS) algorithm is an algorithm that combines the principles of both the Artificial Neural Networks (ANN) self-learning capability

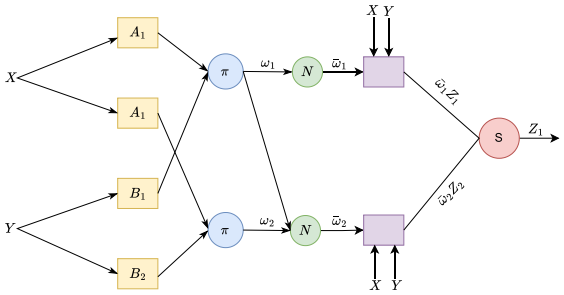


FIGURE 2. CAN bus cyber-attacks detection techniques.

and the fuzzy logic linguistic based models [43]. ANFIS is an algorithm that tunes an initial Fuzzy Inference system (FIS) fuzzy rules using the ANN self-learning capability. ANFIS can also tune the input and output membership functions used to represent the input and output variables through the use of an optimization algorithm. Fig. 2 shows the ANFIS architecture. The ANFIS algorithm has five layers: the fuzzification layer, the product layer, the normalized layer, the de-fuzzification layer, and the output layer. The first layer is for fuzzifying the inputs (x & y) by determining the suitable membership function values (A & B). The second layer is where the multiplications (π) of the previous layer outputs are done. The output of this layer is the fuzzy weights (ω) of each rule. The following layer is the normalization (N) layer where the fuzzy weights values are normalized ($\overline{\omega}$). The fourth layer represents the output membership functions, where the outputs of this layer ($\overline{\omega}z$). are then passed to the fifth and final layer. The final layer is the output layer and it consists of one node. This node sums (S) its inputs from the previous layer to obtain the ANFIS network output (Z).

B. PROPOSED CAN BUS CYBER-ATTACKS DETECTION AND TESTING METHOD

The CAN bus cyber-attack detection techniques were implemented by importing a public dataset available online of an EV CAN bus under cyber-attacks [44] and then implementing the proposed ANFIS-SC and ANFIS-FCM detection techniques after some data preprocessing and oversampling to detect the cyber-attacks. The public dataset includes timestamp, CAN bus ID, Data Length Code (DLC), and a total of 8 data field values. The public dataset was collected by logging the CAN bus traffic through the OBD-II port from a real vehicle while performing the message injection attacks. The performance of the proposed detection techniques is then compared with different machine learning and deep learning-based detection techniques. Fig. 5 shows the flowchart of the step-by-step procedure of the detection techniques implementation to detect and classify the CAN bus cyber-attacks. First, the dataset of the CAN bus under the cyberattacks was imported. Then, data preprocessing techniques were done to obtain the binary-labeled dataset and the multi-labeled dataset and to prepare the datasets

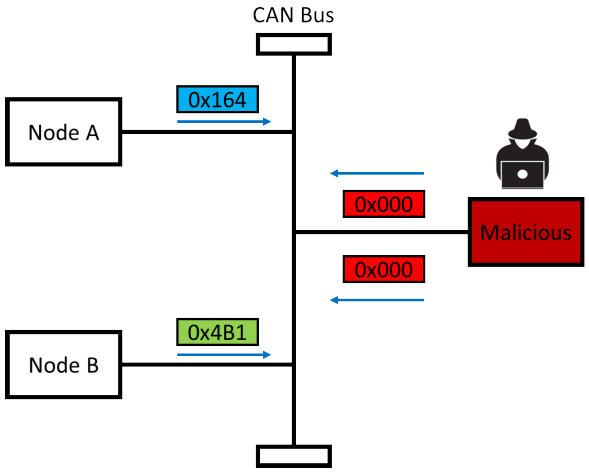


FIGURE 3. DOS attack scenario.

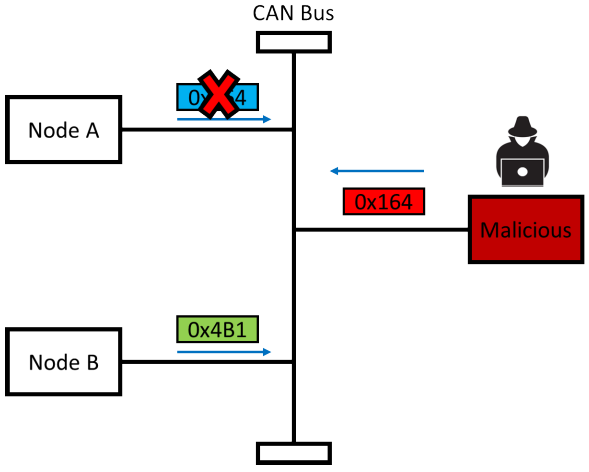


FIGURE 4. Impersonation attack scenario.

for the detection techniques implementation. After that, the preprocessed binary labeled and multi-labeled datasets were split randomly into 80% training and 20% testing datasets. Then, Synthetic The Minority Oversampling Technique (SMOTE) was applied to the preprocessed binary labeled and multi-labeled training datasets to generate more samples of the smaller sample classes and equate them to the majority class. After that, the training datasets post-SMOTE application were used to train the binary and the multiclass classifiers while performing hyper-parameter tuning to optimize the classifiers performances. Next, the trained classifiers were tested on the testing datasets to detect and classify the CAN bus cyber-attacks. The tested binary classifier detects the cyber-attacks and distinguishes whether the CAN bus sample at a certain timestamp is ‘‘normal’’ or under an ‘‘attack’’, while the tested multiclass classifier also detects the cyber-attacks and distinguishes whether the CAN bus sample at a certain timestamp is a ‘‘DOS’’ attack, ‘‘impersonation’’ attack, or ‘‘normal’’. To evaluate the performance of the

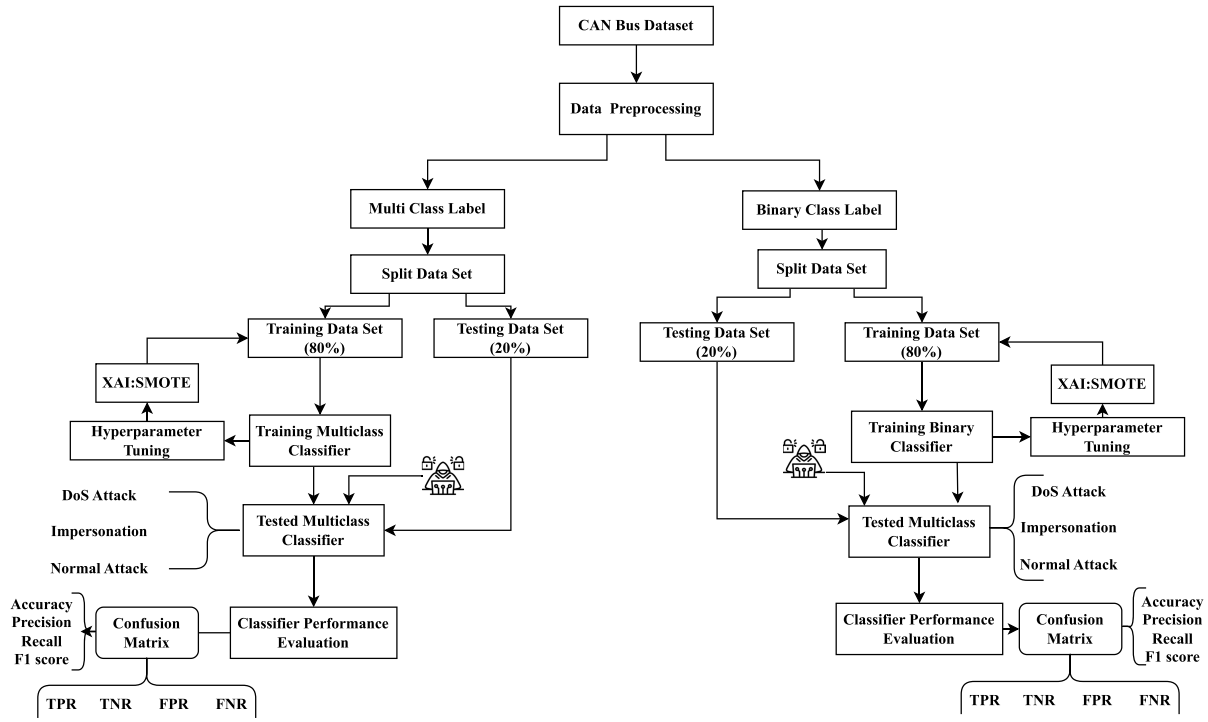


FIGURE 5. CAN bus cyber-attacks detection techniquesimplementation flowchart.

binary classifier and the multiclass classifier, the confusion matrix was used. The testing confusion matrices for the binary and multiclass classifiers were used to evaluate some statistical parameters, which are the True Positive Rate (TPR), True Negative Rate (TNR), False Positive Rate (FPR), False Negative Rate (FNR), Accuracy, Precision, Recall, and the F1 score. These statistical parameters were used to evaluate the performance of the binary and multiclass classifiers in detecting the cyberattacks.

The proposed ANFIS-SC and ANFIS-FCM detection techniques were implemented for both binary classification and multiclass classification of the CAN bus cyber-cyberattacks. Fusion matrices and the evaluated statistical parameters were used to evaluate the detection performances. Machine learning-based detection techniques were implemented for the simple binary classification task, while deep learning-based detection techniques were implemented for the complex multiclass classification task. The performances of the machine learning-based binary classifiers are compared with the performances of the proposed ANFIS-SC and ANFIS-FCM binary classifiers, while the performances of the deep learning-based multiclass classifiers were compared with the performances of the proposed ANFIS-SC and ANFIS-FCM multiclass classifiers. All CAN bus cyber-attack detection techniques are implemented using MATLAB software. The total number of inputs for all techniques is 10, which are the CAN ID, DLC, and the eight data field values, whereas the total number of outputs was two for the binary classification techniques (‘‘Attack’’ and ‘‘Normal classes), and three for the multiclass classification techniques (‘‘DOS’’, ‘‘Impersonation’’, and ‘‘Normal classes).

Data preprocessing is done on the imported dataset to prepare it for the detection technique implementation. A total of 20000 timestamp data points comprising the normal, DOS attack, and impersonation attack classes are imported. The data preprocessing comprises mainly converting the hexadecimal into decimal data points, filling the missing data field values with ‘‘-1’’, and labeling each timestamp sample as either ‘‘Attack’’ or ‘‘Normal’’ for the binary labeled dataset, or as either ‘‘DOS’’, ‘‘Impersonation’’, or ‘‘Normal’’ for the multi-labeled dataset, respectively. The samples of the ‘‘Attack’’ label in the binary labeled dataset comprises the samples of both the ‘‘DOS’’ and ‘‘Impersonation’’ labels in the multi-labeled dataset, while the samples of the ‘‘Normal’’ label are the same for both the binary and multi-labeled datasets. The class labeling is done based on the CAN ID where an ID of ‘‘0 $\times$ 164’’ in hexadecimal format (356 in decimal format) in the imported CAN bus dataset injected with the impersonation attack messages corresponds to an ‘‘Impersonation’’ attack class, an ID of ‘‘0 $\times$ 000’’ corresponds to a ‘‘DOS’’ attack class, and the rest corresponds to the ‘‘Normal’’ or benign class. The number of the ‘‘Normal’’ class samples is much higher than the ‘‘Attack’’ class samples in the binary labeled dataset and is also much higher than the ‘‘DOS’’ and ‘‘Impersonation’’ class samples in the multi-labeled dataset, which results in the so-called class imbalance. Class

imbalance in classification problems is a big factor that can negatively affect the detection techniques' performance where the detection techniques could overlearn the majority classes and underlearn the smaller sample classes and thus have to be solved before the training phase.

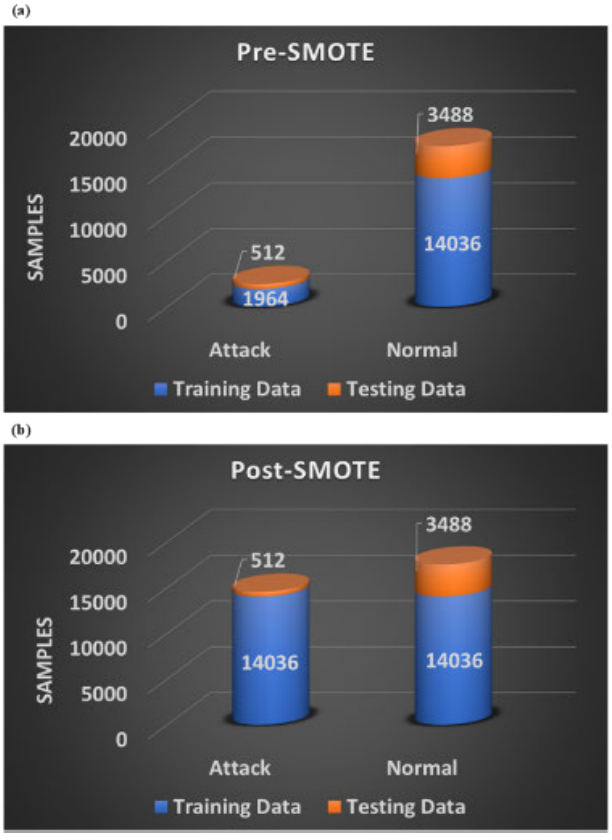


FIGURE 6. Class distribution for the binary labeled training and testing datasets: a) pre-SMOTE application and b) post-SMOTE application.

C. SMOTE IMPLEMENTATION AND CLASS DISTRIBUTION ANALYSIS

SMOTE is a statistical algorithm that can solve the class imbalance problem by synthesizing more samples of the smaller sample classes based on the closest data point samples in the feature space. SMOTE was applied on the preprocessed binary labeled training dataset to generate more samples of the "Attack" class to match it with the "Normal" class and on the multi-labeled training dataset to generate more samples of the "DOS" and "Impersonation" classes to also match them with the "Normal" class. Table 2 shows the SMOTE parameters are used to oversample the training datasets. The number of nearest neighbors to consider for the oversampling is the only parameter of the SMOTE. The value used for this parameter is 30 neighbors to ensure that the smaller sample classes have enough neighbor samples to equate with the majority class. The class distribution before and after applying the SMOTE for the binary-labeled and multi-labeled datasets are shown in Fig. 6 and Fig. 7, respectively. Fig. 6 shows that the class distribution before the SMOTE implementation for the "Attack" and "Normal" classes are 1964 and 14036 for the binary labeled training dataset and is 512 and 3488 for the binary labeled testing dataset, respectively. Fig. 7 shows, on the other hand, that the class distribution before the SMOTE implementation for the "DOS", "Impersonation" and "Normal" classes are 1180, 784, and 14036 for the multi-labeled training dataset and 317, 195, 3488 for the multi labeled testing dataset, respectively. Then, after the implementation of SMOTE, both figures show that all classes had 14036 samples in the training datasets, where the samples in the testing datasets for all classes stayed the same pre and post-SMOTE to ensure the credibility of the obtained testing results of the cyber-attack detection techniques. The post-SMOTE training and testing datasets used were the same for both the proposed binary classifiers and the machine learning-based binary classifiers, and are also the same for both the proposed multiclass classifiers and the deep learning-based multiclass classifiers. This was done to ensure a fair comparison between the performances of the different detection techniques.

TABLE 3. Synthetic minority oversampling technique parameters.

Hyper parameter	Value
Number of nearest neighbours to consider	30

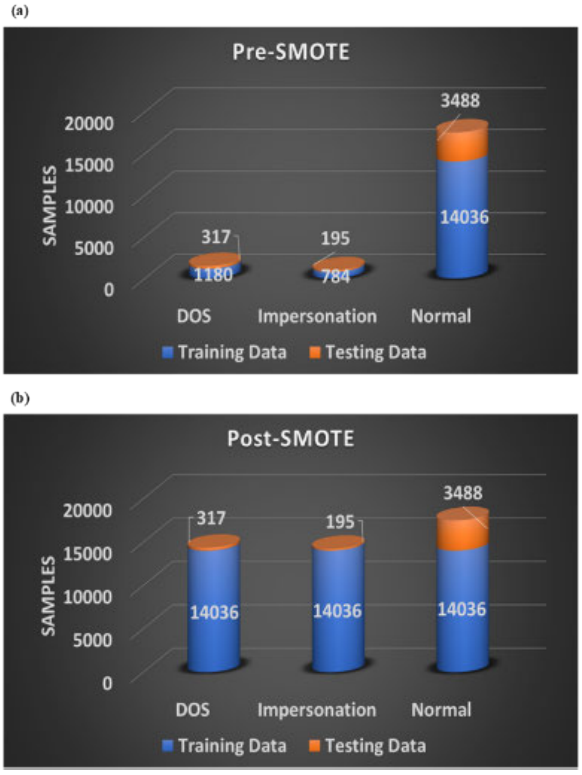


FIGURE 7. Class distribution for the multi-labelled training and testing datasets: a) pre-SMOTE application and b) post-SMOTE application.

D. BINARY AND MULTICLASS CLASSIFIER PERFORMANCE EVALUATION STATISTICAL PARAMETERS

The performance evaluation of the detection techniques is done using different statistical parameters that were evaluated from the testing confusion matrices. The parameters are the TPR, TNR, FPR, FNR, Accuracy, Precision, Recall, and the F1 score. The formulas used to calculate the parameters are as follows [45]:

$$TPR = \frac{TP}{TP + FN} \tag{1}$$

$$TNR = \frac{TN}{TN + FP} \tag{2}$$

$$FPR = \frac{FP}{FP + TN} \tag{3}$$

$$FNR = \frac{FN}{FN + TP} \tag{4}$$

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \tag{5}$$

$$Precision = \frac{TP}{TP + FP} \tag{6}$$

$$Recall = TPR = \frac{TP}{TP + FN} \tag{7}$$

$$F1\ score = 2 \times \frac{Recall \times Precision}{Recall + Precision} \tag{8}$$

where TP, TN, FP and FN stand for ''True Positive'', ''True Negative'', ''False Positive'', and ''False Negative'', respectively.

E. PROPOSED BINARY AND MULTICLASS CLASSIFIERS IMPLEMENTATION: ANFIS-SC AND ANFIS-FCM

This section proposes new CAN bus cyber-attack detection techniques and discusses their implementation procedure in detail. The proposed techniques are ANFIS-SC and ANFIS-FCM, and they were implemented as binary and multiclass classifiers of the CAN bus cyber-attacks. The learning-based ANFIS-FCM is a hybrid combination of ANFIS, which is a deep learning-based algorithm, with the SC and FCM, which are unsupervised machine learning algorithms. ANFIS algorithm is an algorithm that combines the principles of both neural networks and fuzzy logic. The SC and FCM are clustering techniques that can be applied on the input samples to segment them into clusters and assign membership functions with certain values for the purpose of fuzzification. The fuzzified inputs are then processed through the subsequent layers of the ANFIS algorithm to finally be defuzzified at the output layer to output a crisp value. The fuzzy logic incorporated in the ANFIS algorithm enables the algorithm to be flexible and to be used for a variety of applications. The proposed techniques were implemented using the binary and multi-labeled post-SMOTE training and testing datasets, where the binary and multi-labeled training datasets were used for training the proposed binary classifiers and multiclass classifiers, respectively. The binary-labeled and multi-labeled testing datasets were used for testing the performance of the proposed detection techniques.

The ANFIS architecture illustrated in Fig. 2 shows that the algorithm has multiple layers with only one output node. The output node for an ANFIS used as a classifier, can be interpreted as the probability that an input sample belongs to the positive class. This means that while the ANFIS algorithm can be used for binary classification problems since there is only one positive class and one negative class, it cannot be used for multiclass classification problems since there are multiple positive and negative classes. Although the ANFIS algorithm can be used for binary classification problems, its performance may not be the best. This demands the development of a framework that enables the implementation of the ANFIS algorithm for classification problems with good performance. The developed framework for the proposed binary and multiclass classifiers was based on the idea that one ANFIS-SC or ANFIS-FCM algorithm is used to detect each class for the binary and multiclass classifications. For each algorithm in the binary and multiclass classifications, the target class that the algorithm wants to predict is the positive class, while the classes are the negative classes. This is achieved through the one-hot encoding of the ground truth or the target class labels in the training datasets to obtain vectors with the value of ''1'' for the positive class and the value of ''0'' for the negative classes for each sample. This method will enable each ANFIS-SC or ANFS-FCM algorithm to act as binary detectors for the corresponding positive class. The one-hot encoded vectors of each class for the binary classification are ''Attack'' = {1, 0} and ''Normal'' = {0, 1}, and for the multiclass classification, the vectors of each class are ''DOS'' = {1, 0, 0}, ''Impersonation'' = {0, 1, 0}, and ''Normal'' = {0, 0, 1}. In total, we will have two one-hot encoded vectors for the binary classification training dataset and three hot encoded vectors for the multiclass classification training dataset. The length of the vectors for the binary and multi-labeled training datasets is the same as the number of samples of the binary and multi-labeled training datasets. Each vector of the training datasets was used to train one ANFIS-SC or ANFIS-FCM classifier, and then the trained classifiers were used on the testing datasets to output the probabilities of each sample belonging to the positive class. To improve the classifiers' performances, an augmented ANFIS-SC or ANFIS-FCM classifier was added after the binary and multiclass classifiers. The augmented classifiers were passed the output probabilities of the preceding classifiers as inputs, then output the class predictions of each sample. The augmented classifiers were trained after the training of the preceding classifiers, receiving their training output probabilities as inputs and learning to match the class predictions of each sample to the ground truth or target class labels in the training datasets. Then, during the testing phase, the trained augmented classifiers were also tested after the testing of the preceding classifiers, where

they received their testing output probabilities as inputs to output the class predictions of each sample. In total, there are three ANFIS-SC or ANFIS-FCM classifiers for the binary classification and four ANFIS-SC or ANFIS-FCM classifiers for the multiclass classification.

The SC, FCM, and ANFIS hyperparameter values for the binary and multiclass classifications are listed in Table 3. The SC and FCM hyperparameters control the number of clusters and their centers in addition to controlling the number of fuzzy rules in the ANFIS algorithm. The hyperparameter values were selected based on a trial and error process where a range of values for each hyperparameter were used for training the ANFIS-SC and ANFIS-FCM algorithms using the training datasets and then evaluating the performances of the trained algorithms using the testing datasets. The initial Fuzzy Inference System (FIS) for the ANFIS algorithms can be generated using either SC or FCM. The optimization algorithm is a hybrid algorithm that combines backpropagation and least squares estimation. The number of epochs needed to train the algorithms was only 1, since additional epochs only increased the training time without improving the performance of the algorithms.

F. MACHINE LEARNING BASED BINARY CLASSIFIERS AND DEEP LEARNING-BASED MULTICLASS CLASSIFIERS IMPLEMENTATION

This section discusses the implementation procedure for machine learning based binary classifiers and deep learning-based multiclass classifiers with their hyperparameter tuning. The post-SMOTE binary labeled and labeled training datasets were used to train the machine learning based binary classifiers and the deep learning-based multiclass classifiers, respectively. Then, the trained machine learning-based binary classifiers and deep learning-based multiclass classifiers were tested on the binary-labeled and multi-labeled testing datasets, respectively, to evaluate their performances. The implemented machine learning binary classifiers are some of the standard machine-learning algorithms, and they are Linear Discriminant Analysis (LDA), Logistic Regression (LR), Support Vector Machines (SVM), and Random Forest (RF). On the other hand, the implemented deep learning-based multiclass classifiers are state-of-the-art deep learning-based detection techniques in the literature, and they are Convolutional Neural Network (CNN), Convolutional Neural Network-Long Short-Term Memory Network (CNN-LSTM), Gated Recurrent Unit (GRU) and Long Short-Term Memory (LSTM) networks. The hyperparameter values of the machine learning-based binary classifiers and the deep learning-based multiclass classifiers were selected based on a trial and error process where a range of values for each hyperparameter was used for training the machine learning and deep learning-based algorithms using the training datasets and then evaluating the performances of the trained algorithms using the testing datasets.

IV. EXPERIMENTAL RESULTS

A. BINARY CLASSIFICATION RESULTS

This section presents and discusses the binary classification training and testing results of the proposed ANFIS-SC and ANFIS-FCM detection techniques in addition to the standard machine learning-based detection techniques. The performance evaluation of the binary classifiers was done using the training and testing confusion matrices and the computed TPR, TNR, FPR, FNR, accuracy, precision, recall, and F1 score values from the testing confusion matrices. The confusion matrices were used to evaluate the performance of the detection techniques since the problem that the implemented CAN bus cyber-attack detection techniques are trying to solve a classification problem.

TABLE 4. Subtractive clustering, fuzzy C-Means clustering, and ANFIS hyperparameters values.

	Hyper parameter	Value
Subtractive Clustering	Cluster range of influence	0.20 (binary) & 0.25 (multi)
	Squash factor	0.20 (binary) & 0.25 (multi)
	Accept ratio	0.5
	Reject ratio	0.15
Fuzzy C-Means Clustering	Number of clusters	10
	Exponent	2
	Maximum number of iterations	100
	Distance metric	Euclidean
ANFIS	Fuzzy inference system generation method	SC or FCM
	Optimization algorithm	Hybrid
	Number of epochs	1

TABLE 5. Binary classification performances of the proposed ANFIS-SC and ANFIS-FCM (True & False).

Algorithm	Accuracy	Class	TPR	TNR	FPR	FNR
ANFIS-SC	**99.58%**	Attack	1	0.995	0.005	0
		Normal	0.995	1	0	0.005
		Overall	**0.998**	**0.998**	**0.002**	**0.002**
ANFIS-FCM	**99.68%**	Attack	1	0.996	0.004	0
		Normal	0.996	1	0	0.004
		Overall	**0.998**	**0.998**	**0.002**	**0.002**

TABLE 6. Comparison between the binary classification performances of the proposed ANFIS-SC and ANFIS-FCM.

Algorithm	Accuracy	Class	Precision	Recall	F1 score
Linear		Attack	0.476	1	0.645
Discriminant	**85.93%**	Normal	1	0.839	0.912
Analysis		**Overall**	**0.738**	**0.919**	**0.819**
Logistic		Attack	0.592	1	0.744
Regression	**91.18%**	Normal	1	0.899	0.947
		Overall	**0.796**	**0.949**	**0.866**
		Attack	0.837	1	0.911
SVM	**97.50%**	Normal	1	0.971	0.986
		Overall	**0.918**	**0.986**	**0.951**
		Attack	0.946	1	0.973
Random Forest	**99.28%**	Normal	1	0.992	0.996
		Overall	**0.973**	**0.996**	**0.984**
		Attack	0.968	1	0.984
ANFIS-SC	**99.58%**	Normal	1	0.995	0.998
		Overall	**0.984**	**0.998**	**0.991**
		Attack	0.975	1	0.988
ANFIS-FCM	**99.68%**	Normal	1	0.996	0.998
		Overall	**0.988**	**0.998**	**0.993**

The binary classification testing performances of the proposed ANFIS-SC and ANFIS-FCM detection techniques and

the comparison of their testing performances with standard machine learning-based detection techniques are depicted in Tables 4 and 5, respectively. The binary classification testing performances were evaluated using different statistical parameters, which are the TPR, TNR, FPR, FNR, accuracy, precision, recall, and F1 score values. These statistical parameters were computed based on the testing confusion matrices of the binary classifiers. A perfect binary classifier will have the values of 1.0 (100%) for the accuracy, precision, recall, F1 score, TPR, and TNR parameters, and the values of 0 (0%) for the FPR and FNR parameters. Table 4 shows the testing accuracy values and the TPR, TNR, FPR, and FNR values of each class and the overall average values for the proposed ANFIS-SC and ANFIS-FCM binary classifiers. The testing accuracies of the proposed ANFIS-SC and ANFIS-FCM binary classifiers were 99.58% and 99.68%. The overall average values of TPR, TNR, FPR, and FNR were 0.9976, 0.9976, 0.0024, and 0.0024 for the proposed ANFIS-SC binary classifier and 0.9981, 0.9981, 0.0019, and 0.0019 for the proposed ANFIS-FCM binary classifier, respectively. The high testing accuracy, TPR and TNR values, in addition to the low FPR and FNR values of the proposed ANFIS-SC and ANFIS-FCM binary classifiers prove that the proposed detection techniques were successful in the detection of CAN bus cyber-attacks. The ANFIS-FCM binary classifier slightly outperformed the ANFIS-SC binary classifier since it had slightly higher testing accuracy, TPR, and TNR values, in addition to slightly lower FPR and FNR values. Table 5 shows the accuracy, precision, recall, and F1 score values for each class and the overall average values for the proposed detection techniques and for standard machine learning-based detection techniques to compare their performances. The testing accuracy values for the LDA, LR, SVM, RF, ANFIS-SC, and ANFIS-FCM binary classifiers were 85.93%, 91.18%, 97.50%, 99.28%, 99.58%, and 99.68%, respectively, while the overall average F1 score values were 0.8188, 0.8659, 0.9508, 0.9844, 0.9907, and 0.9928, respectively. The F1 score value relies on the precision and recall values, which makes it the suitable parameter for comparing the performances of the detection techniques alongside the testing accuracies. The results show that the proposed ANFIS-SC and ANFIS-FCM detection techniques have higher testing accuracies and overall F1 scores and thus have better performances compared to standard machine learning-based detection techniques.

B. MULTICLASS CLASSIFICATION RESULTS

This section presents and discusses the multiclass classification training and testing results of the proposed ANFIS-SC and ANFIS-FCM detection techniques in addition to state-of-the-art deep learning-based detection techniques in the literature. The performance evaluation of the multiclass classifiers is done using the training and testing confusion matrices and the computed TPR, TNR, FPR, FNR, accuracy, precision, recall, and F1 score values from the testing confusion matrices. The confusion matrices were used to evaluate the performance of the detection techniques since the problem that the implemented CAN bus cyber-attack detection techniques are trying to solve a classification problem.

TABLE 7. Multiclass classification performances (true & false).

Algorithm	Accuracy	Class	TPR	TNR	FPR	FNR
ANFIS-SC	**99.65%**	DOS	1	1	0	0
		Imp	1	0.997	0.003	0
		Normal	0.996	1	0	0.004
		Overall	**0.999**	**0.999**	**0.001**	**0.001**
ANFIS-FCM	**99.68%**	DOS	1	1	0	0
		Imp	1	0.997	0.003	0
		Normal	0.996	1	0	0.004
		Overall	**0.999**	**0.999**	**0.001**	**0.001**

TABLE 8. Multiclass classification performances.

Algorithm	Accuracy	Class	Precision	Recall	F1 score
CNN [46]	**82.63%**	DOS	0.317	0.316	0.316
		Imp	0.177	0.328	0.230
		Normal	0.945	0.901	0.922
		Overall	**0.480**	**0.515**	**0.497**
CNN-LSTM [47]	**86.45%**	DOS	0.383	0.344	0.362
		Imp	0.167	0.005	0.01
		Normal	0.903	0.960	0.930
		Overall	**0.484**	**0.436**	**0.459**
GRU [48]	**99.50%**	DOS	1	1	1
		Imp	0.940	0.959	0.949
		Normal	0.998	0.997	0.997
		Overall	**0.979**	**0.985**	**0.982**
LSTM [7]	**99.55%**	DOS	1	1	1
		Imp	0.932	0.980	0.955
		Normal	0.999	0.996	0.997
		Overall	**0.977**	**0.992**	**0.984**
ANFIS-SC	**99.65%**	DOS	0.997	1	0.998
		Imp	0.938	1	0.968
		Normal	1	0.996	0.998
		Overall	**0.978**	**0.999**	**0.988**
ANFIS-FCM	**99.68%**	DOS	1	1	1
		Imp	0.938	1	0.968
		Normal	1	0.996	0.998
		Overall	**0.979**	**0.999**	**0.989**

The multiclass classification testing performances of the proposed ANFIS-SC and ANFIS-FCM detection techniques and the comparison of their testing performances with the state-of-the-art deep learning-based detection techniques in the literature are depicted in Tables 6 and 7, respectively. The dataset comprised three different classes which were DOS, Impersonation (Imp), and Normal. The multi-class classification testing performances were evaluated using different statistical parameters, which are the TPR, TNR, FPR, FNR, accuracy, precision, recall, and F1 score values. These statistical parameters were computed based on the testing confusion matrices of the multiclass classifiers. A the perfect multiclass classifier will have values of 1.0 (100%) for the accuracy, precision, recall, F1 score, TPR, and TNR parameters and the values of 0 (0%) for the FPR and FNR parameters. Table 6 shows the testing accuracy values and the TPR, TNR, FPR and FNR values of each class and the overall average values for the proposed ANFIS-SC and ANFIS-FCM multiclass classifiers. The testing accuracies of the proposed

TABLE 9. Training and testing computational time.

Algorithm	Training	Testing
NN [46]	04 min 49 s	0.200 s
GRU [48]	43 min 41 s	0.415 s
LSTM [7]	45 min 52 s	0.467 s
CNN-LSTM [47]	18 min 51 s	0.381 s
ANFIS-SC	13 min 08 s	0.150 s
ANFIS-FCM	01 min 16 s	0.210 s

ANFIS-SC and ANFIS-FCM multiclass classifiers were 99.65% and 99.68%. The overall average values of TPR, TNR, FPR, and FNR were 0.9987, 0.9988, 0.0012, and 0.0013 for the proposed ANFIS-SC multiclass classifier and 0.9988, 0.9989, 0.0011, and 0.0012 for the proposed ANFIS-FCM multiclass classifier, respectively. The high testing accuracy, TPR, and TNR values, in addition to the low FPR and FNR values of the proposed ANFIS-SC and ANFIS-FCM multiclass classifiers, prove that the proposed detection techniques were successful in the detection of CAN bus cyberattacks. The ANFIS-FCM multiclass classifier slightly outperformed the ANFIS-SC multiclass classifier since it had slightly higher testing accuracy, TPR, and TNR values, in addition to slightly lower FPR and FNR values. Table 7 shows the accuracy, precision, recall, and F1 score values for each class and the overall average values for the proposed detection techniques and for the state of the art of deep learning-based detection techniques in the literature to compare between their performances. The testing accuracy values for the CNN, CNN-LSTM, GRU, LSTM, ANFIS-SC, and ANFIS-FCM multiclass classifiers were 82.63%, 86.45%, 99.50%, 99.55%, 99.65%, and 99.68%, respectively, while the overall average F1 score values were 0.4966, 0.4589, 0.9821, 0.9843, 0.9883, and 0.9889, respectively. The F1 score value relies on the precision and recall values, which makes it the suitable parameter for comparing the performances of the detection techniques alongside the testing accuracies. The results show that the proposed ANFIS-SC and ANFIS-FCM detection techniques have higher testing accuracies and overall F1 scores and thus have better performance compared to the state-of-the-art deep learning-based detection techniques in the literature.

The comparison between the training and testing computational times of the proposed ANFIS-SC and ANFIS-FCM detection techniques with state-of-the-art deep learning-based detection techniques are shown in Table 8. The training and testing of the proposed techniques for multiclass classification were done on MATLAB, which is run on an Intel(R) Core (TM) i5-10210U CPU @1.60GHz and 8.00 GB RAM. The training of the state ofstate-of-the-artrning-based detection techniques was also done on MATLAB but using an NVIDIA GeForce MX230 with 2.0 GB RAM, whereas the testing was done on the Intel(R) Core (TM) i5-10210U CPU @1.60GHz and 8.00 GB RAM. The SMOTE implementation on the multi-labeled dataset took 0.0515 seconds. Table 8 shows that the training times of state-of-the-art deep learning-based detection techniques are below 46 minutes, while their testing times are below 0.5 seconds. The table also shows that the training and testing times of the proposed ANFIS-SC and ANFIS-FCM detection techniques are below 14 minutes and below 0.25 seconds, respectively. The proposed detection techniques had lower testing times than state-of-the-art deep learning-based detection techniques, except for the CNN where they were approximately the same. This is due to implementing the fuzzy logic inside the ANFIS-SC and ANFIS-FCM, which speeds up the computation process. Although the testing times of the proposed detection techniques and the CNN detection technique are approximately the same, the CNN detection technique had a very bad performance. The results of the multiclass classification detection techniques show that the proposed techniques had the highest performance and the lowest testing time compared to the state-of-the-art deep learning-based detection techniques, and thus, they are the most suitable for the CAN bus cyber-attacks multiclass classification application.

V. CONCLUSION AND FUTURE WORK

The aim of this research was to develop and propose an EV CAN bus cyber-attacks detection technique based on the Adaptive Neuro-Fuzzy Inference System (ANFIS) algorithm and test their performances in detecting and classifying the cyber-attacks. New detection techniques based on the combination of the ANFIS algorithm with SC and FCM (ANFIS-SC and ANFIS-FCM) were proposed and implemented. Then, a comparison between the ANFIS algorithm-based detection techniques performances with the other developed machine learning and deep learning-based techniques were done. The high testing accuracy (∼99.6%), TPR and TNR values (above 99.8%) in addition to the low FPR and FNR values (less than 0.2%) of the proposed ANFIS-SC and ANFIS-FCM detection techniques, proved that the proposed detection techniques were successful in the detection of CAN bus cyber-attacks. The proposed ANFIS-SC and ANFIS-FCM detection techniques had higher testing accuracies (∼99.6%) and overall F1 scores (above 98.8%) and thus had better performance compared to standard machine learning-based techniques and the state-of-the-art deep learning-based detection techniques. Future research directions include implementing metaheuristic algorithms such as Particle Swarm Optimization, as optimization algorithms for the proposed detection techniques to improve their performances.

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